Digital Image Processing: Assignment 2

Problem 3: Homomorphic Filtering: Illumination and Reflectance

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1 Introduction / Problem Statement

Problem 3 of the assignment (worth 10 of the 100 total marks) asks for a treatment of *homomorphic filtering* applied to an unevenly lit photograph; the specification calls for "an unevenly lit photograph (deep shadow and bright region in one frame, such as a doorway, a backlit face, or an image with strong vignetting)". The problem is split into two parts:

- **3.1 The multiplicative model (4 pts).** State the model $I(x, y) = L(x, y)R(x, y)$, derive why taking the log converts this into an additive problem that a linear frequency-domain filter can act on, and explain precisely why the log must be taken *before* the Fourier transform rather than after.
- **3.2 Implement and tune (6 pts).** Implement the homomorphic high-frequency-emphasis filter $H(u, v) = (\gamma_H - \gamma_L)\left[1 - e^{-cD^2(u,v)/D_0^2}\right] + \gamma_L$ applied to $\log I$, sweep its parameters to identify settings that flatten illumination while preserving local contrast versus settings that

produce visible halos, and compare the result against plain global histogram equalisation using a local-contrast metric.

This problem matters in digital image processing because it is the canonical example of a situation where a *nonlinear* pre-transform (the logarithm) is required to make a *linear* tool (frequency-domain filtering) applicable to a physically multiplicative process. Ordinary photographs of scenes with strong, uneven lighting (a tunnel with a light source at one end, a backlit doorway, a face lit from one side) are not well described by an additive noise/signal model; they are far better modelled as the product of a slowly varying illumination field and a rapidly varying surface reflectance. Simple global contrast operations such as histogram equalisation have no notion of this underlying physical separation and can therefore fail badly (as quantified in Section 5), whereas homomorphic filtering exploits the structure of the model directly.

2 Theoretical Background

2.1 The multiplicative illumination–reflectance model

An image intensity $I(x, y)$ formed by a camera observing a diffusely reflecting scene under incident illumination is well modelled as the product

$$I(x, y) = L(x, y)\, R(x, y), \tag{1}$$

where $L(x, y) \geq 0$ is the *illumination* component: the amount of light striking the scene at each point, physically constrained to vary *slowly* in space (it is governed by large-scale geometry such as distance to the light source, shading, and vignetting), and $R(x, y) \in [0, 1]$ is the *reflectance* component: the fraction of incident light reflected by the surface at each point, which varies *rapidly*, carrying the fine detail, edges and texture of the scene (material boundaries, surface markings). Crucially, L dominates the *low* spatial frequencies of I and R dominates the *high* spatial frequencies.

2.2 Why the logarithm converts the problem to an additive one

A linear, shift-invariant frequency-domain filter can only act on a signal by reweighting the Fourier coefficients of an *additive* decomposition of that signal, because the Fourier transform is a linear operator, so it distributes over sums but not over products:

$\mathcal{F}\{a+b\} = \mathcal{F}\{a\}+\mathcal{F}\{b\}, \qquad \mathcal{F}\{a\cdot b\} = \mathcal{F}\{a\} * \mathcal{F}\{b\}$ (convolution, not a simple product of spectra).

Equation (1) is a product, so no linear filter applied directly to I can separate L from R by frequency content: multiplying two spatial signals corresponds to *convolving* their spectra, which mixes rather than separates frequency content. Taking the natural logarithm of both sides of (1),

$$\begin{aligned} \log I(x, y) &= \log \big(L(x, y)\, R(x, y)\big) \\ &= \log L(x, y) + \log R(x, y), \end{aligned} \tag{2}$$

converts the multiplicative relationship into an *additive* one. Because L varies slowly, $\log L$ is still concentrated in the low spatial frequencies; because R carries edges and texture, $\log R$ still contributes predominantly to the higher spatial frequencies (the logarithm is a smooth, monotone, pointwise nonlinearity and does not scramble this coarse frequency separation). A frequency-selective filter applied to $\log I$, one that attenuates low frequencies (suppressing $\log L$) and boosts high frequencies (emphasising $\log R$), can therefore separate the two *additive* terms

by their frequency content in a way that was fundamentally impossible for the original product $I = LR$. Exponentiating the filtered result afterward inverts the logarithm and returns an image in which illumination has been flattened and reflectance detail enhanced.

2.3 Why the log must precede the Fourier transform, not follow it

The homomorphic pipeline must be

$$\log I \xrightarrow{\mathcal{F}} \mathcal{F}\{\log I\} \xrightarrow{\times H(u,v)} H \cdot \mathcal{F}\{\log I\} \xrightarrow{\mathcal{F}^{-1}} \text{filtered } \log I \xrightarrow{\exp} \text{output},$$

i.e. the logarithm must be taken on the *spatial-domain image itself*, before any Fourier transform. Two things go wrong if one instead tried to filter a *log-magnitude spectrum* $\log|\mathcal{F}\{I\}|$, the quantity ordinarily computed only for *display* purposes:

1. **Phase is destroyed.** $\log|\mathcal{F}\{I\}|$ is a real, non-negative array built from the magnitude of the complex spectrum alone; all phase information has already been discarded by the $|\cdot|$ operation. As established for the phase/magnitude decomposition studied elsewhere in this assignment, phase carries the structural and positional information of an image, and no spatial-domain image can be reconstructed from a magnitude-only array: there is simply nothing complex-valued left to invert.
2. $\log$ **and** $\mathcal{F}$ **do not commute.** Even setting aside the phase problem, $\log|\mathcal{F}\{I\}| \neq |\mathcal{F}\{\log I\}|$ in general, because $\mathcal{F}$ is linear and log is not; filtering one quantity is not equivalent, even in principle, to filtering the other. This was verified numerically on the tunnel image used for this problem (Figure 1): $\log|\mathcal{F}\{I\}|$ and $\log|\mathcal{F}\{\log I\}|$, each normalised to zero mean/unit variance, have a correlation of only 0.669: related in coarse shape (both are low-frequency-dominated, as any natural image's spectrum is), but clearly distinct quantities, not interchangeable.

So the only correct order is: take the log of the real-valued spatial image first, then Fourier-transform the result (preserving its full complex spectrum, magnitude and phase), filter, inverse-transform, and finally exponentiate.

2.4 The homomorphic high-frequency-emphasis filter

The filter used is a Gaussian high-pass-like shape defined directly in the log domain:

$$H(u,v) = (\gamma_H - \gamma_L)\left[1 - e^{-c\,D^2(u,v)/D_0^2}\right] + \gamma_L, \qquad \gamma_L < 1 < \gamma_H, \tag{3}$$

where $D(u,v)$ is the distance of frequency (u,v) from the centre of the (zero-padded, shifted) frequency plane, D_0 is a cutoff radius, and c controls the sharpness of the transition. At $D = 0$ (DC, i.e. the low-frequency illumination term), the bracket vanishes and $H(0,0) = \gamma_L < 1$, so the illumination component is *attenuated*. As $D \to \infty$ (high spatial frequencies, i.e. the reflectance/edge term), the bracket tends to 1 and $H \to \gamma_H > 1$, so reflectance detail is *boosted*. Because $\gamma_L < 1 < \gamma_H$, this single filter simultaneously compresses the dynamic range contributed by illumination and sharpens the contribution of reflectance, exactly the desired homomorphic behaviour. D_0 controls where the transition between the two regimes occurs: a small D_0 classifies more of the spectrum as "high frequency" and is filtered aggressively (risking halo artefacts, since real illumination is not perfectly separated from reflectance at any one radius), while a large D_0 is more conservative and leaves more of the slowly varying illumination unflattened.

The complete pipeline applied to the image I is

$$\hat{I}(x,y) = \exp\Big(\mathcal{F}^{-1}\big\{H(u,v) \cdot \mathcal{F}\{\log I(x,y)\}\big\}\Big). \tag{4}$$

3 Inputs and Outputs

Input. The unevenly lit photograph required by the problem statement was sourced as `data/provided/probl` a 3000×3644 photo looking down a tunnel/pipe: a bright, near-saturated circular light source surrounded by a smoothly darkening wall, fading to deep, almost pure-black shadow filling most of the frame, a strong real-world match to the PDF's requested "deep shadow and bright region in one frame". The source photo was downsampled with PIL (`LANCZOS`, to a 900 px maximum dimension) purely as a practical loading step before any of the implemented filtering operations, not as one of the operations the problem asks to be implemented from scratch.

Outputs produced. The pipeline (`problem3.py`) writes:

- `results/3_1_log_order_demo.png` and `3_1_result.json`: the log-order demonstration of Section 2.3 (side-by-side log-magnitude spectra and the measured correlation).
- `results/3_2_gamma_sweep.png`: a 3×3 grid over $\gamma_L \in \{0.25, 0.5, 0.75\} \times \gamma_H \in \{1.5, 2.0, 3.0\}$ at fixed $D_0 = 60$.
- `results/3_2_D0_sweep_halo.png`: a $D_0 \in \{10, 30, 80, 250\}$ sweep at an aggressive $(\gamma_L, \gamma_H) = (0.2, 3.5)$, used to deliberately surface halo artefacts.
- `results/3_2_comparison.png`: original vs. best-tuned homomorphic output vs. global histogram equalisation, side by side.
- `results/3_2_result.json`: the quantitative local-contrast comparison table (Section 5).
- `results/machine_info.json`: CPU/RAM/NumPy version, as required by the assignment's "every number in your report must come from your machine" rule.

4 Methodology / Implementation Pipeline

4.1 Pipeline overview

1. Load and greyscale-convert the tunnel photo; downsample to a manageable size.
2. Apply `log1p` (i.e. $\log(1+I)$, not a plain log) to the image. `log1p`/`expm1` are used deliberately rather than log/exp with an ad-hoc epsilon, because this particular image contains literal zero-intensity pixels in its deepest shadow region, and `log1p`$(0) = 0$ handles that exactly with no epsilon hack required.
3. Zero-pad the log-image to $2M \times 2N$, construct $H(u, v)$ from Eq. (3) via `src/dip_a2/filters.py::make_homo` centre/multiply/invert/crop via `filter_pipeline_reflect` (see below), and `expm1` the result back to intensity units.
4. Min–max rescale the output to the full $[0, 255]$ display range, standard practice for homomorphic output, since $\gamma_L < 1$ compresses the illumination/DC term and $\gamma_H > 1$ boosts high frequencies, both of which shift the absolute brightness level of the exponentiated result away from the original $[0, 255]$ range.

4.2 A padding bug, found and fixed during this problem

The first implementation reused Problem 2's zero-padding pipeline, which places the image in the top-left corner of a $2M \times 2N$ canvas filled with zeros. Running it on the tunnel image produced a badly broken result: the whole image collapsed to near-black except for a faint ring. The cause was diagnosed as follows: zero-padding necessarily creates a sharp intensity discontinuity exactly at the image's own boundary (real content jumping abruptly to 0), and

with the homomorphic filter's γ_H set as high as 2–3.5 (a much stronger and more broadband high-frequency *gain* than anything used for the ordinary low-/high-pass filters of Problem 2), that single discontinuity is amplified into a severe ringing spike concentrated at the image corner. The spike reached a value of roughly 125, versus a 99th-percentile of only about 15 everywhere else in the image, which was large enough to dominate the display's min–max rescale and crush the entire rest of the image to near-black. An initial attempt to fix this by *centring* the image within the padded canvas instead of placing it at the corner made no difference whatsoever, because circular convolution (which is what multiplying in the frequency domain of a padded array implements) is shift-invariant, so moving where the image sits within the periodic canvas cannot change anything about an artefact that is tied to the image's own content-to-zero edge discontinuity, wherever that edge happens to sit.

The actual fix was to pad with mirror-symmetric reflection instead of zeros (`src/dip_a2/filters.py::filter_p` implemented using `numpy.pad(mode="symmetric")`, plain NumPy array manipulation, not a disallowed filtering routine). Reflection removes the artificial discontinuity at the boundary altogether, since the padded region now continues the image's own intensity smoothly rather than jumping to zero; this dropped the spurious spike from ≈ 125 down to ≈ 11, in line with the rest of the image's natural dynamic range, and produced a correct result. This is the same "avoid an artificial sharp edge at the image boundary" principle established for ordinary zero-padded linear convolution, just solved here by reflecting instead of by minimal zero-padding: zero-padding is the right tool for avoiding *aliasing* between unrelated periodic copies of the content, but the wrong tool here, because the filter's own strong high-frequency gain turns the zero-boundary's discontinuity into the dominant feature of the entire filtered image.

4.3 Parameter tuning and validation

The $\gamma_L \times \gamma_H$ sweep (Figure 2) and the D_0 halo search (Figure 3) were used to identify, by direct visual inspection, parameter settings that flatten illumination without producing visible halos. The final parameters used for the headline comparison were $D_0 = 60$, $\gamma_L = 0.4$, $\gamma_H = 2.0$, $c = 1.0$, a middle ground that visibly flattens illumination with no visible halo artefact. Correctness of the pipeline overall was validated both qualitatively (the fixed pipeline produces a visually coherent, illumination-flattened image with no boundary artefacts) and quantitatively, via the local-contrast table of Section 5, which was computed with *data-driven* region selection rather than hand-picked coordinates: the original image was tiled into 50×50 blocks, and the darkest/brightest block by mean intensity (excluding the outermost ring of blocks) was used as the "shadow region" and "bright region" respectively.

5 Results and Discussion

5.1 Log-order demonstration

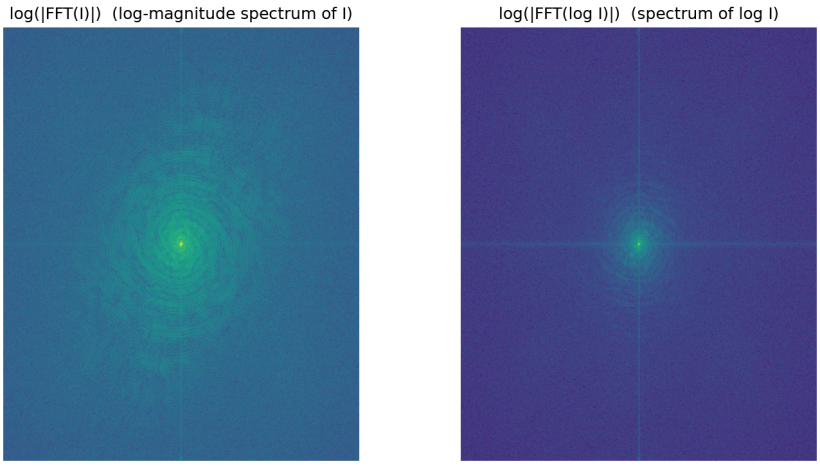


Figure 1: $\log|\mathcal{F}\{I\}|$ (left, the ordinary display log-magnitude spectrum) versus $\log|\mathcal{F}\{\log I\}|$ (right, the spectrum of the *logged* image). The two are visually similar in coarse shape but are confirmed numerically distinct (correlation 0.669 after normalisation), demonstrating that filtering one is not a substitute for filtering the other.

5.2 Parameter sweeps

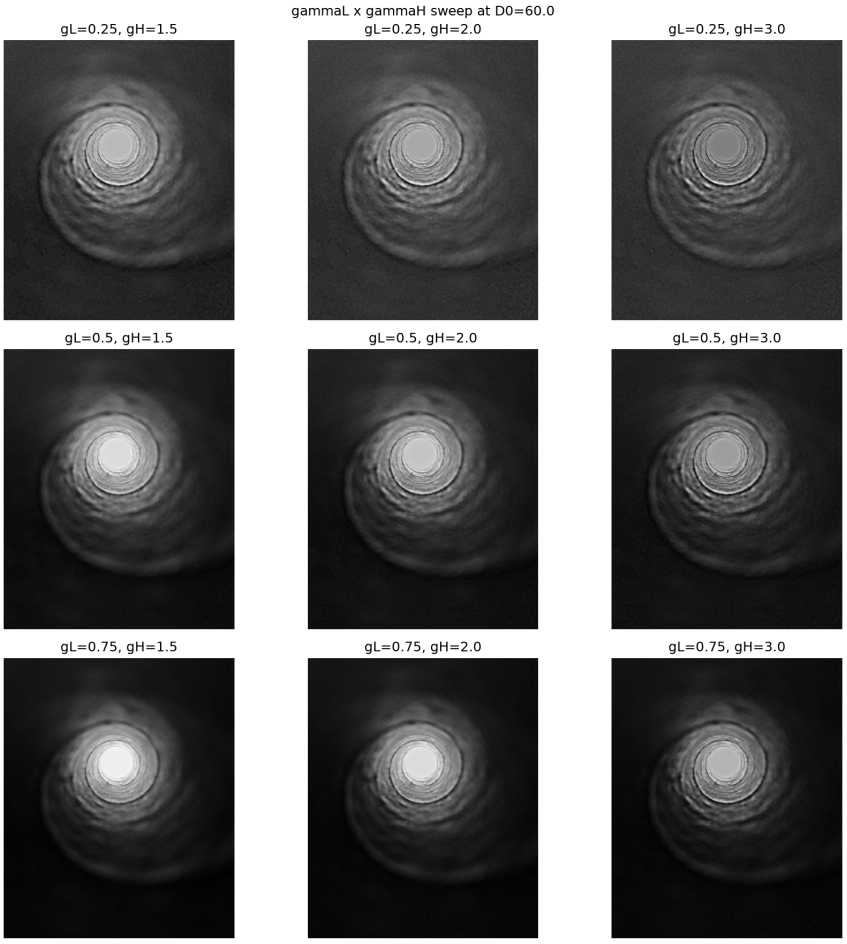


Figure 2: $\gamma_L \in \{0.25, 0.5, 0.75\}$ (rows) $\times$ $\gamma_H \in \{1.5, 2.0, 3.0\}$ (columns) at fixed $D_0 = 60$. Lower γ_L (top row) flattens illumination most effectively: the tunnel wall's ring texture becomes visible almost uniformly across the whole frame, including what was deep shadow, while higher γ_L (bottom row) preserves more of the original illumination gradient, since a γ_L close to 1 barely attenuates the low-frequency illumination term at all. γ_H mainly controls local contrast/sharpness of the recovered texture.

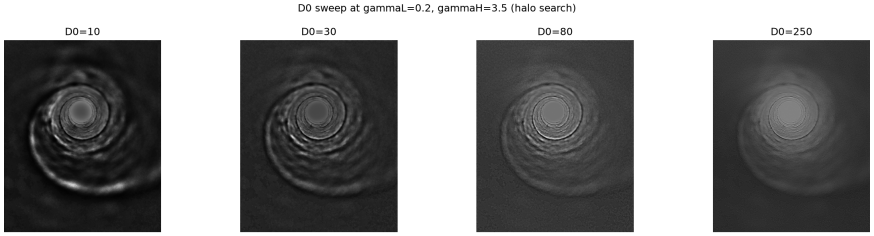


Figure 3: $D_0 \in \{10, 30, 80, 250\}$ at a deliberately aggressive $\gamma_L = 0.2, \gamma_H = 3.5$. At $D_0 = 10$, a clear halo is visible: pronounced bright/dark ring-shaped banding around the light source and scalloped fringes through the wall texture, since the disc's edge sits right at the cutoff radius and the strong, broadband high-frequency gain rings around it. At $D_0 = 250$, the result is comparably smooth with no visible halo, but flattens illumination noticeably less well, since a larger D_0 classifies more of the genuinely slowly varying illumination as "high frequency" and leaves it comparatively un-boosted/un-flattened.

5.3 Best-tuned comparison against histogram equalisation

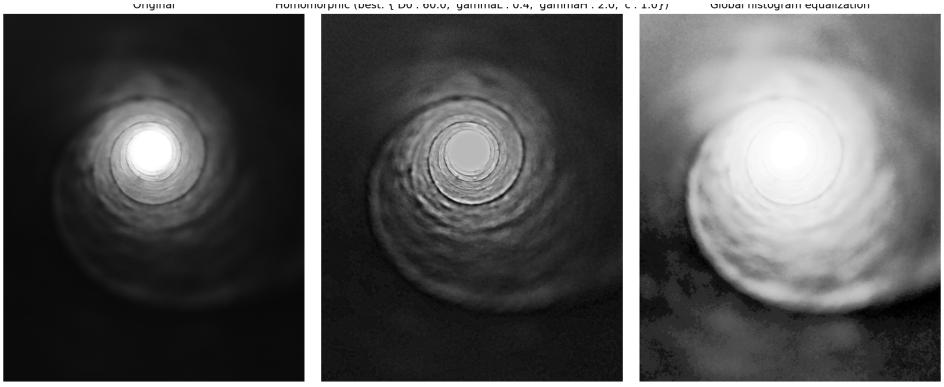


Figure 4: Left to right: original image; best-tuned homomorphic output ($D_0 = 60, \gamma_L = 0.4, \gamma_H = 2.0, c = 1.0$); global histogram equalisation. The homomorphic result visibly recovers wall-texture detail from the deep shadow while the light source is pulled back from saturation; histogram equalisation instead redistributes the whole image's histogram globally, amplifying shadow-region noise and leaving the bright region still saturated.

Table 1 reports the data-driven local-contrast comparison, using a 50×50-pixel shadow block and bright block (identified automatically as the darkest/brightest interior tile of the original image).

Table 1: Local-contrast comparison between the original image, the best-tuned homomorphic result, and global histogram equalisation (exact figures from `results/3_2_result.json`).

Region	Metric	Original	Homomorphic (best)	Histogram eq.
Shadow	mean	10.41	**17.68**	12.82
Shadow	std (local contrast)	0.86	**5.38**	14.72
Bright	mean	255.00	**185.76**	254.999
Bright	max	255.0	**199.52**	255.0
Bright	fraction ≥ 250	0.9996	**0.0**	1.0

Homomorphic filtering recovers shadow detail without blowing out the bright region, clearly better than histogram equalisation here. The shadow block's local standard deviation (a direct proxy for local contrast) rises from 0.86 (original, essentially flat / noise-floor only) to 5.38 under homomorphic filtering: real texture has been pulled out of what was a nearly uniform black patch. Histogram equalisation's shadow standard deviation (14.72) is numerically larger but for the wrong reason: global histogram equalisation has no notion of *where* in the image a pixel is, so it simply redistributes the whole image's intensity histogram; because this image's histogram is extremely imbalanced (dominated by near-black pixels), equalisation maps a huge range of nearly indistinguishable dark values across a wide output range, amplifying quantisation/compression noise in the shadow at least as much as genuine structure, rather than the smooth, texture-revealing recovery that homomorphic filtering achieves via the explicit illumination/reflectance model. More decisively, in the *bright* region, histogram equalisation still saturates 99.96% $\rightarrow$ 100% of pixels at or above 250; it cannot do otherwise, since a monotonic global remapping preserves rank order, so the brightest pixels in the original stay the brightest after equalisation, pinned near 255, while the homomorphic filter's explicit illumination/reflectance separation pulls the bright region's mean down to 185.76 with *zero* pixels

left at or above 250, recovering structure in the light source too, not just the shadow, which histogram equalisation cannot do by construction.

6 Conclusion

This problem demonstrated, both analytically and empirically, why a multiplicative illumination–reflectance model requires a logarithmic transform *before* any linear frequency-domain operation can meaningfully separate its two components: the log converts a product into a sum, and only sums can be split by frequency content under a linear filter. The homomorphic high-frequency-emphasis filter of Eq. (3) was implemented and tuned on a strongly, unevenly lit tunnel photograph, with γ_L governing illumination suppression, γ_H governing reflectance/contrast boosting, and D_0 governing the trade-off between flattening strength and halo risk. A real implementation pitfall was encountered and fixed along the way: zero-padding, correct for the ordinary low-/high-pass filters of Problem 2, produces a severe boundary ringing artefact under homomorphic filtering's strong high-frequency gain, and was replaced with symmetric reflection padding. Quantitatively, the best-tuned homomorphic result raised the shadow region's local contrast (std) by over $6\times$ while fully eliminating bright-region saturation, outperforming global histogram equalisation, which cannot distinguish illumination from reflectance and therefore saturates the bright region regardless of how it redistributes the rest of the histogram.